GGG–C–430a
March 24, 1967
SUPERSEDING
Int. Fed. Spec. GGG–C–00430 (GSA–FSS)
June 23, 1964

FEDERAL SPECIFICATION

CLEANER, VALVE STEM GUIDE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers the requirements for one type of valve stem guide cleaner for removing carbon from valve guides in internal combustion engines.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Federal Standards:

Fed. Std. No. 123—*Marking for Domestic Shipment* (Civilian Agencies).
Fed. Test Method Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL–H–15424—Hand Tools; Packaging of.

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5110

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:39Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (NBS) Handbook:
H28—Screw-Thread Standards for Federal Services.

(Copies may be obtained upon application, accompanied by money order, coupon, or cash to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402. Price may be obtained from the Superintendent of Documents.)

3. REQUIREMENTS

3.1 Illustration. The illustration shown herein is for the convenience of identification, and is not intended to preclude the purchase of valve guide cleaners which are otherwise in accordance with the requirements of this specification.

3.2 Material. The material used in the construction of the valve stem guide cleaner shall be as specified herein. All materials shall be of uniform quality, and free from any defects or imperfections that would affect the serviceability of the tool.

3.3 Design. The valve guide cleaner shall consist essentially of a shaft threaded on one end, blades, 2 cups, and a threaded nut. When assembled, the tool shall be similar to figure 1. The tool shall have a minimum of four blades and shall be adjustable from 5/16 to 9/16 inch minimum diameter, at the cutting edge. Adjustment shall be achieved through the threaded nut on the shaft. Screw threads shall be in conformance with the current issue of the NBS H28. At the option of the contractor, the cleaner shall be provided with a handle for hand operation.

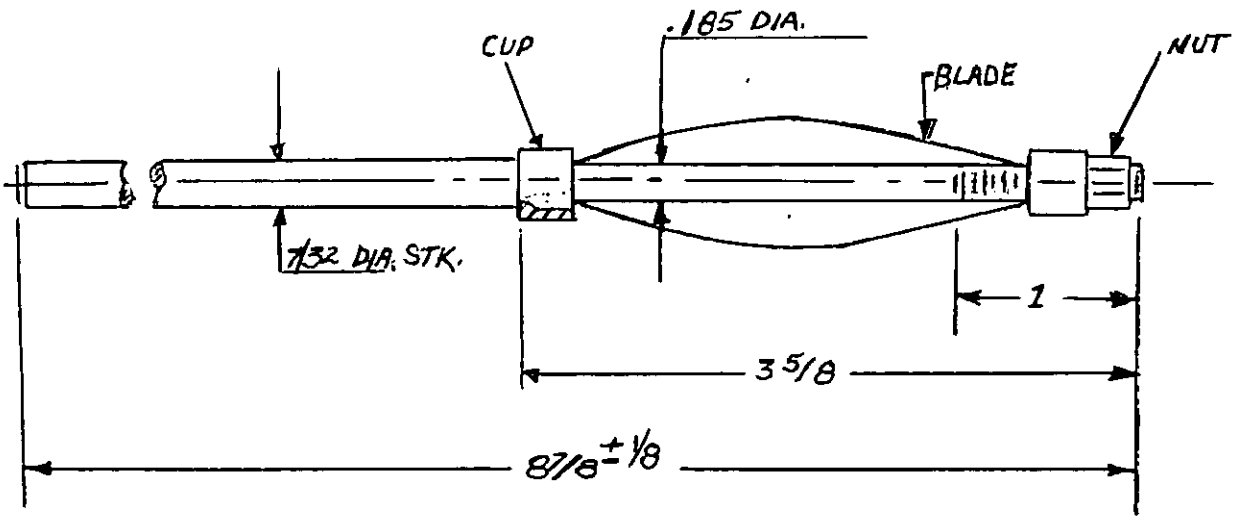


Figure 1. Cleaner, valve stem guide.

3.4 Shaft, cups, and nut. The shaft, cups, and nut shall be constructed of a steel possessing a minimum carbon content of 0.18 percent. Dimensions shall be as specified in figure 1.

3.5 Blades. Blades shall be of spring steel, 3 inches long, plus or minus ¼ inch. The width of the blade shall be 0.125 ± 0.010 inch. The thickness of the blade shall be 0.020 ± 0.010 inch. One extra set of blades, of the quantity normally assembled on the cleaner shall be furnished with the tool.

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:39Z
Check the source to verify that this is the current version before use.

3.6 Hardness. The hardness of the cleaner blades shall be not less than Rockwell C 45 nor more than C 58.

3.7 Finish. All surfaces shall be free from burrs, cracks, scale, and other detrimental defects.

3.8 Identification. The cleaner shall be marked in a permanent and legible manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.9 Workmanship. Workmanship shall be in accordance with high grade industrial practice.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL–STD–105. Data for sampling shall be stated in table I.

Table I. Sampling data.

Category	Sample unit	Inspection level	Acceptable quality level	AQL expressed in terms of	Reference
Visual examination	1 ea.	II	2.5	Percent defective	4.3.1
Dimensional examination	1 ea.	II	2.5	do.	4.3.2
Testing	1 ea.	S–1	1.5	do.	4.4
Preparation for delivery	One container	S–2	4.0	Defects per hundred units	4.6

4.3 Examination.

4.3.1 Visual examination. Each sample unit shall be examined for non-conformance in design, material, finish, coating, construction, workmanship, and marking.

4.3.2 Dimensional examination. Each sample unit shall be examined for any nonconformance with dimensional requirements.

4.4 Testing. Each sample unit shall be tested in accordance with 4.5.

4.5 Hardness test. The blades of each sample cleaner shall be tested in accordance with method 243.1 of Fed. Test Method Std. No. 151 to determine compliance with the Rockwell hardness specified in 3.6.

4.6 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing and marking comply with the requirements in section 5. Defects shall be scored in accordance with table II. The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot.

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:39Z
Check the source to verify that this is the currentversion before use.

Table II. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure container flaps, loose strapping, inadequate stapling.
	Distortion of container.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packing shall be level A or C, as specified (see 6.1).

5.1.1 Level A. The cleaner shall be preserved and packaged in accordance with MIL–H–15424, as specified for level A.

5.1.2 Level C. The cleaner shall be preserved and packaged in accordance with the supplier's commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. The cleaners shall be packed in accordance with MIL–H–15424, as specified for level A.

5.2.2 Level B. The cleaners shall be packed in accordance with MIL–H–15424, as specified for level B.

5.2.3 Level C. The cleaners shall be packed to insure carrier acceptance and safe delivery at destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the unit packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to markings required by the contract or order, the unit packages and shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following data in procurement documents:

(a) Title, number, and date of this specification.

(b) Selection of applicable levels of preservation, packaging, and packing required (see 5.1 and 5.2).

DOD coordination has been waived.

Preparing activity:

GSA—FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 5 cent each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:39Z
Check the source to verify that this is the current version before use.

★ GPO : 1987 O - 352-512 (320)